Lab-9 Kalman Filter

Aim:
To implement a Kalman Filter in python to estimate room temperature by combining the previous estimate with sensor readings and updating the temperature estimate and uncertainity after each reading.

Step 1: Initialise the temperature estimate
x = 20.0
p = 4.0
x stores the initial temperature estimate as 20°c
p stores the uncertainity variance as $4°C^2$.

Step2: Set the noise values
Q = 1.0
R = 4.0
Q represents uncertainity due to temperature changes
R represents uncertainity in the thermometer readings.

Step 3: Store the sensor readings
readings = [22.0, 21.0, 23.0]
The thermometer gives 3 readings : 22°c, 21°c & 23°c

Step 4: Process each reading
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for step in range( len (readings)):
    z = readings [step]
```
The loop processes each reading one by one.
step takes the values 0, 1 and 2
z stores the current sensor reading.

Step 5: Predict the Temperature

$x_pred = x$

$P_pred = P + Q$

x_pred stores the predicted temperature, assuming no known temperature trend.

P_pred stores the predicted variance, which increases by Q.

Initially, x_pred = 20 and P_pred = 4+1 = 5.

Step 6: Calculate Kalman Gain

$K = P_pred / (P_pred + R)$

K determines how much the sensor readings influence the estimate.

For the first readings, $K = 5 / (5+4) = 0.5556$

Step 7: Calculate the Difference

$difference = z - x_pred$

This calculates the difference between the sensor reading and predicted temperature.

For the first reading $22 - 20 = 2°C$

Step 8: Correct the temperature estimate

$correction = K * difference$

$x = x_pred + correction$

correction calculates how much to adjust the prediction

The corrected estimate is $20 + (0.5556 \times 2) \approx 21.111°C$

Step 9: Update the uncertainity

$P = (1 - K) * P_pred$

This updates the variance after considering the sensor reading.

For the first reading, $P \approx (1 - 0.5556) \times 5 = 2.22°C^2$

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Step 10: Display the result

print ("Reading", step + 1, "sensor", z,
"estimate", round (x, 4), "K", round (K, 4),
"Variance", round (P, 4))

step +1 displays the reading number starting from 1.
round() displays values to four decimal places.
The output shows the sensor reading, corrected estimate, Kalman gain, and updated variance.

Initial values $x = 20$, $P = 4$, $Q = 1$, $R = 4$

Reading 1: $z = 22$

Variable	Calculation	Value
x_pred	x	20.0000
P_pred	P + Q = 4 + 1	5.0000
K	5 / (5 + 4)	0.5556
difference	22 - 20	2.0000
correction	0.5556 x 2	1.1111
x	20 + 1.1111	21.1111
P	(1 - 0.5556) x 5	2.2222

Reading 2: $z = 21$

Variable	Calculation	Value
x_pred	x	21.1111
P_pred	2.2222 + 1	3.2222
K	3.2222 / (3.2222 + 4)	0.4459
difference	21 - 21.1111	-0.1111
correction	0.4459 x (-0.1111)	-0.0496
x	21.1111 - 0.0496	21.0615
P	(1 - 0.4459) x 3.2222	1.7850

Reading : 3 $z = 23$

Variable	Calculation	Value
x_pred	x	21.0615
P_pred	$1.7850 + 1$	2.7850
K	$2.7850 / (2.7850 + 4)$	0.4105
difference	$23 - 21.0615$	1.9385
correction	0.4105×1.9385	0.7959
x	$21.0615 + 0.7959$	21.8574
P	$(1 - 0.4105) \times 2.7850$	1.6416

Reading 1 sensor 22.0 estimate 21.1111, K 0.5556, variance 2.2222

Reading 2 sensor 21.0 estimate 21.0615, K 0.6459, variance 1.7850

Reading 3 sensor 23.0 estimate 21.8574, K 0.4105 variance 1.6416